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Inventor(s)

Determan; Alexander R. et al.

VALVE WITH IMPROVED MAGNETIC FLUX INTERFACE

Abstract

A valve can include a valve body, a plunger within the valve body, and a flux collar coupled to the valve body. A distance between the flux collar and the plunger can vary around an inner perimeter of the flux collar. A port section, a flux collar support section, and a coil support section of the valve body can be a unitary structure, such as an injection molded structure. The flux collar support section of the valve body can include an annular wall and/or a plurality of longitudinal ribs. The inner perimeter of the flux collar can be an annular inner wall and/or can have a plurality of longitudinal grooves spaced to receive the ribs.

Inventors: Determan; Alexander R. (Grand Blanc, MI), Hoskins; Matthew Tyler (Oak Park, MI), Ward; James R. (Sturgis, MI)

Applicant: ASCO, L.P. (Florham Park, NJ)

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS

[0001] This application is related to the commonly owned applications entitled “SOLENOID VALVE WITH BARBED SEALING ASSEMBLY,” “SOLENOID PLUNGER ALIGNMENT,” “SOLENOID VALVE POPPET AFFIXMENT,” and “PRESS TO SPECIFICATION SOLENOID VALVE ASSEMBLY,” respectively, each filed on the same day as the instant application, and the contents of which are incorporated herein by reference.

BACKGROUND OF THE INVENTION

[0002] Field of the Invention. The present disclosure relates generally to solenoid valves and more specifically relates to miniature solenoid valves.

Description of the Related Art.

[0003] Fundamentally, all electromagnetic solenoid devices turn electrical energy into mechanical work. One of the characteristics that defines how efficient a solenoid is at converting electrical energy into mechanical work is the series magnetic airgap within the solenoid. Reducing the series magnetic airgap increases the efficiency of the system. Many solenoids have some sort of flux collar or soft magnetic pathway to reduce the series magnetic airgap within the solenoid. However, conventional designs have limits.

SUMMARY OF THE INVENTION

[0004] Applicants have created new and useful devices, systems and methods for solenoid valves. In at least one embodiment, the series magnetic airgap can be minimized by thinning and/or eliminating a wall section between a flux collar and a plunger of the solenoid valve. In at least one embodiment, the series magnetic airgap can be minimized while the valve body, including that section between the flux collar and the plunger, nonetheless remains strong enough to sufficiently resist bending and axial loading during assembly and operation of the valve. In at least one embodiment, the series magnetic airgap can be minimized while the valve body, including that section between the flux collar and the plunger, nonetheless remains strong enough to sufficiently resist the pressure of media within the valve and/or is capable of being formed via injection molding (e.g., of plastic). In at least one embodiment, the series magnetic airgap can be minimized and the valve body, such as the section between the flux collar and the plunger, can be advantageously configured for supporting formation via injection molding and avoiding constrictions that can be problematic during the injection molding process.

[0005] In at least one embodiment, a valve can include a valve body, a plunger within the valve body, a flux collar coupled to the valve body, or any combination thereof. In at least one embodiment, a distance between the flux collar and the plunger can vary around an inner perimeter of the flux collar. In at least one embodiment, the valve body can include a port section, a flux collar support section, and a coil support section extending along a longitudinal axis. In at least one embodiment, the port section, the flux collar support section, and the coil support section of the valve body can be or include a single structure, such as a unitary or monolithic injection molded structure.

[0006] In at least one embodiment, the plunger can extend at least partially through the flux collar support section and the coil support section. In at least one embodiment, the flux collar can be coupled to the flux collar support section of the valve body. In at least one embodiment, the plunger can extend at least partially through the flux collar.

[0007] In at least one embodiment, the flux collar support section of the valve body can include an annular wall and/or a plurality of longitudinal ribs. In at least one embodiment, the ribs can extend outwardly from the annular wall of the flux collar support section of the valve body. In at least one embodiment, the inner perimeter of the flux collar can be an annular inner wall and/or can have a plurality of longitudinal grooves spaced to receive the ribs.

[0008] In at least one embodiment, the flux collar can be a multi-section ring. In at least one embodiment, each section of the flux collar can have at least one longitudinal groove configured to

receive one of the ribs. For example, the flux collar can be a two-section ring. In at least one embodiment, the sections of the flux collar can interface along a plane. In at least one embodiment, the ribs can extend from the annular wall of the flux collar support section of the valve body, such as along one or more axes perpendicular to the plane.

[0009] In at least one embodiment, the flux collar support section of the valve body can have an annular inner wall and an outer wall. In at least one embodiment, the outer wall of the flux collar support section can have one or more flat sides and/or one or more lobes. In at least one embodiment, the inner perimeter of the flux collar can be an inner wall having at least two flat sides corresponding to the flat sides of the outer wall of the flux collar support section of the valve body. In at least one embodiment, the inner perimeter of the flux collar can be an inner wall having a contour corresponding to, or matching, the outer wall of the flux collar support section of the valve body.

[0010] In at least one embodiment, the flux collar support section of the valve body can have a plurality of longitudinal ribs, with or without an annular wall. In at least one embodiment, the inner perimeter of the flux collar can be an annular inner wall having a plurality of longitudinal grooves spaced to receive the ribs. In at least one embodiment, the inner wall of the flux collar can be flush with an inner wall of the coil support section of the valve body.

[0011] In at least one embodiment, the coil support section of the valve body includes an annular wall having a constant thickness. In at least one embodiment, the flux collar support section of the valve body includes an annular wall having a variable thickness, which can vary around the wall. In at least one embodiment, the variable thickness of the flux collar support section can be less than the constant thickness of the coil support section around at least a portion of the flux collar support section of the valve body.

[0012] In at least one embodiment, the flux collar support section of the valve body can have a variable thickness, that varies around the flux collar support section of the valve body. In at least one embodiment, the coil support section of the valve body can have a constant thickness. In at least one embodiment, the variable thickness of the flux collar support section of the valve body can be thinner than the constant thickness of the coil support section of the valve body around at least a portion of the flux collar support section. In at least one embodiment, the variable thickness of the flux collar support section of the valve body can be thicker than the constant thickness of the coil support section of the valve body around at least a portion of the flux collar support section.

[0013] In at least one embodiment, valve body can include a port section, a flux collar support section, and a coil support section extending along a longitudinal axis. In at least one embodiment, the port section, the flux collar support section, and the coil support section of the valve body can be or include a unitary structure, such as an integrally formed, injection molded structure. In at least one embodiment, the flux collar support section of the valve body can have a variable thickness, that varies around the flux collar support section of the valve body.

[0014] In at least one embodiment, the coil support section of the valve body can have a constant thickness around the coil support section of the valve body. In at least one embodiment, the variable thickness of the flux collar support section of the valve body can be thinner than the thickness of the coil support section of the valve body around at least a portion of the flux collar support section. In at least one embodiment, the variable thickness of the flux collar support section of the valve body can be thicker than the thickness of the coil support section of the valve body around at least a portion of the flux collar support section.

Description

BRIEF DESCRIPTION OF DRAWINGS

[0015] FIG. 1 is a sectional view of one of many embodiments of a solenoid valve according to the

disclosure.

[0016] FIG. 2 is a partial perspective view of one of many embodiments of a valve body according to the disclosure.

[0017] FIG. 3 is a cross-sectional view of one of many embodiments of a solenoid valve having a flux collar according to the disclosure.

[0018] FIG. 4 is a cross-sectional view of another one of many embodiments of a solenoid valve having a flux collar according to the disclosure.

[0019] FIG. 5 is a cross-sectional view of yet another one of many embodiments of a solenoid valve having a flux collar according to the disclosure.

[0020] FIG. 6 is a cross-sectional view of still another one of many embodiments of a solenoid valve having a flux collar according to the disclosure.

[0021] FIG. 7 is a perspective view of one of many embodiments of a valve body having a flux collar according to the disclosure.

[0022] FIG. 8 is a perspective view of one of many embodiments of a flux collar according to the disclosure.

DETAILED DESCRIPTION OF THE INVENTION

[0023] The figures described above and the written description of specific structures and functions below are not presented to limit the scope of what Applicants have invented or the scope of the appended claims. Rather, the figures and written description are provided to teach any person skilled in the art to make and use the inventions for which patent protection is sought. Those skilled in the art will appreciate that not all features of a commercial embodiment of the inventions are described or shown for the sake of clarity and understanding. Persons of skill in this art will also appreciate that the development of an actual commercial embodiment incorporating aspects of the present inventions will require numerous implementation-specific decisions to achieve the developer's ultimate goal for the commercial embodiment. Such implementation-specific decisions may include, and likely are not limited to, compliance with system-related, business-related, government-related and other constraints, which may vary by specific implementation, location and from time to time. While a developer's efforts might be complex and time-consuming in an absolute sense, such efforts would be, nevertheless, a routine undertaking for those of skill in this art having benefit of this disclosure. It must be understood that the inventions disclosed and taught herein are susceptible to numerous and various modifications and alternative forms.

[0024] The use of a singular term, such as, but not limited to, "a," is not intended as limiting of the number of items. Also, the use of relational terms, such as, but not limited to, "top," "bottom," "left," "right," "upper," "lower," "down," "up," "side," and the like are used in the written description for clarity in specific reference to the figures and are not intended to limit the scope of the inventions or the appended claims. The terms "including" and "such as" are illustrative and not limitative. The terms "couple," "coupled," "coupling," "coupler," and like terms are used broadly herein and can include any method or device for securing, binding, bonding, fastening, attaching, joining, inserting therein, forming thereon or therein, communicating, or otherwise associating, for example, mechanically, magnetically, electrically, chemically, operably, directly or indirectly with intermediate elements, one or more pieces of members together and can further include without limitation integrally forming one functional member with another in a unity fashion. The coupling can occur in any direction, including rotationally. Further, all parts and components of the disclosure that are capable of being physically embodied inherently include imaginary and real characteristics regardless of whether such characteristics are expressly described herein, including but not limited to characteristics such as axes, ends, inner and outer surfaces, interior spaces, tops, bottoms, sides, boundaries, dimensions (e.g., height, length, width, thickness), mass, weight, volume and density, among others.

[0025] Any process flowcharts discussed herein illustrate the operation of possible implementations of systems, methods, and computer program products according to various embodiments of the

present disclosure. In this regard, each block in a flowchart may represent a module, segment, or portion of code, which can comprise one or more executable instructions for implementing the specified logical function(s). It should also be noted that, in some implementations, the function(s) noted in the block(s) might occur out of the order depicted in the figures. For example, blocks shown in succession may, in fact, be executed substantially concurrently. It will also be noted that each block of flowchart illustration can be implemented by special purpose hardware-based systems that perform the specified functions or acts, or combinations of special purpose hardware and computer instructions.

[0026] Applicants have created new and useful devices, systems and methods for solenoid valves, such as miniature solenoid valves. Once again, reducing the series magnetic airgap of a solenoid valve can increase the efficiency of the valve. In at least one embodiment, the series magnetic airgap can be minimized by thinning and/or eliminating a wall section between a flux collar and a plunger of the solenoid valve. In at least one embodiment, the series magnetic airgap can be minimized while the valve body, including that section between the flux collar and the plunger, nonetheless remains strong enough to sufficiently resist bending and axial loading during assembly and operation of the valve. In at least one embodiment, the series magnetic airgap can be minimized while the valve body, including that section between the flux collar and the plunger, nonetheless remains strong enough to sufficiently resist the pressure of media within the valve and/or is capable of being formed via injection molding (e.g., of plastic). In at least one embodiment, the series magnetic airgap can be minimized and the valve body, such as the section between the flux collar and the plunger, can be advantageously configured for supporting formation via injection molding and avoiding constrictions that can be problematic during the injection molding process.

[0027] In at least one embodiment, the valve body, such as the section between the flux collar and the plunger, can include a wall having a set of ribs, such as a plurality of longitudinal ribs, equally or otherwise spaced about the circumference or perimetry thereof. In at least one embodiment, the ribs can increase the moment of inertia in a wall or wall section for providing strength to the valve body for resisting applicable loads while simultaneously providing for a relatively small series magnetic airgap between the flux collar and the plunger. In at least one embodiment, one or more ribs can be advantageously configured for supporting sufficient material flow during injection molding, i.e., to sufficiently fill the valve body mold during manufacturing. In at least one embodiment, one or more ribs can be advantageously configured for allowing thinner wall sections in at least a portion of the valve body versus conventional valve bodies, thereby allowing for the flux collar to be closer in position to the plunger of the solenoid, minimizing the average airgap, and increasing efficiency of the solenoid.

[0028] FIG. 1 is a sectional view of one of many embodiments of a solenoid valve according to the disclosure. FIG. 2 is a partial perspective view of one of many embodiments of a valve body according to the disclosure. FIG. 3 is a cross-sectional view of one of many embodiments of a solenoid valve having a flux collar according to the disclosure. FIG. 4 is a cross-sectional view of another one of many embodiments of a solenoid valve having a flux collar according to the disclosure. FIG. 5 is a cross-sectional view of yet another one of many embodiments of a solenoid valve having a flux collar according to the disclosure. FIG. 6 is a cross-sectional view of still another one of many embodiments of a solenoid valve having a flux collar according to the disclosure. FIG. 7 is a perspective view of one of many embodiments of a valve body having a flux collar according to the disclosure. FIG. 8 is a perspective view of one of many embodiments of a flux collar according to the disclosure. FIGS. 1-8 are described in conjunction with one another.

[0029] In at least one embodiment, a valve **100** according to the disclosure, such as a miniature solenoid valve, can include one or more valve bodies **200**, one or more plungers **300** disposed at least partially within the valve body **200**, and one or more flux collars **400** configured to be coupled to the valve body **200** for supporting operation of the valve. In at least one embodiment, a valve

100 can include a shell **150** for housing or protecting one or more other components of the valve. In at least one embodiment, a distance or gap between the flux collar **400** and the plunger **300** can vary along at least a portion of the valve body **200**, such as around or about an inner perimeter or perimetry of the flux collar **400**. In at least one embodiment, a wall thickness of at least a portion of the valve body **200** can vary along at least a portion of the valve body **200**, such as a portion disposed radially between an outer surface of the plunger **300** and an inner surface of the flux collar **400**.

[0030] In at least one embodiment, the valve body **200** can include one or more port sections **210**, one or more flux collar support sections **220**, one or more coil support sections **230**, or any combination thereof, extending along one or more axes, such as a central longitudinal axis X. In at least one embodiment, the port section **210**, the flux collar support section **220**, and the coil support section **230** of the valve body **200** can be unitary or monolithic, such as by way of being portions of an integrally formed structure (e.g., an injection molded structure). In at least one embodiment, the port section **210** can include one or more common ports **212**, one or more normally open ports **214**, one or more normally closed ports **216**, or any combination thereof.

[0031] In at least one embodiment, the plunger **300** can extend at least partially through the flux collar support section **220** and the coil support section **230**. In at least one embodiment, the plunger **300** can extend at least partially through the flux collar **400**. In at least one embodiment, the plunger **300** can include one or more necks **310** that extend into the port section **210**. In at least one embodiment, the plunger **300** can include one or more poppets **320** mounted to the neck **310**. In at least one embodiment, one or more biasing members, such as a spring **330**, can bias the plunger **300** and/or poppet **320** towards or away from one or more seats **218**. In at least one embodiment, a poppet **320** can include a plurality of components, such as a poppet body and one or more seals or other components coupled thereto. In at least one embodiment, a poppet **320** can be a unitary or monolithic structure, such as, for example, a solid elastomer structure.

[0032] In at least one embodiment, the coil support section **230** of the valve body **200** can include one or more annular walls **232** and/or one or more coils **234**. In at least one embodiment, the wall **232** can support coil **234**, which can selectively actuate, or move, the plunger **300** against the force of the spring **330**, to thereby switch the valve **100** from a normally open position to a normally closed position. In at least one embodiment, the coil support section **230** can include a core **238** and one or more stops **236** to limit the movement of the plunger **300** in one or more directions within the bore **240**. As shown in FIG. 1 for illustrative purposes, in at least one embodiment, the stop **236** and the core **238** can be parts of a unitary structure, such as an injection molded structure. As another example, in at least one embodiment, the stop **236** and the core **238** can be two or more separate parts coupled to one another and/or to other components of valve **100** as needed or desired in accordance with an implementation of the disclosure.

[0033] In at least one embodiment, the flux collar **400** can be coupled to the flux collar support section **220** of the valve body **200**, such as by way of being disposed about a wall or other portion of the flux collar support section **220**. In at least one embodiment, the flux collar **400** can be disposed at least partially in a groove **242** of the flux collar support section **220**, such as a groove **242** located between two or more retainers **244**, such as walls, stops, collars or other structures. In at least one embodiment, the flux collar support section **220** of the valve body **200** can include one or more walls **222**, such as an annular wall about a bore **240** in which plunger **300** may be disposed, and/or one or more ribs **224**. In at least one embodiment, the ribs **224** can extend radially outwardly from the wall **222** of the flux collar support section **220** of the valve body **200**, such as from an outside or exterior surface of the wall **222**. In at least one embodiment, one or more of the ribs **224** can extend outwardly along an axis perpendicular to an exterior surface of the wall **222** of the flux collar support section **220** of the valve body **200**. In at least one embodiment, the inner perimeter of the flux collar **400** can include one or more inner walls **402** and/or one or more longitudinal grooves **404** spaced to receive at least a portion of one or more of the ribs **224**.

[0034] In at least one embodiment, the flux collar **400** can be or include a multi-section ring, such as a ring comprising a plurality of segments or other portions. In at least one embodiment, each section **410** of the flux collar **400** can have one or more longitudinal grooves **404** configured to receive or otherwise cooperate with at least a portion of one or more corresponding ribs **224**. For example, in at least one embodiment, the flux collar **400** can be a two-section ring. In at least one embodiment, the sections **410** of the flux collar **400** can meet, separate, or otherwise interface at or along a plane. In at least one embodiment, the ribs **224** can extend from the wall **222** of the flux collar support section **220** of the valve body **200**, such as radially outwardly or in a radially outward direction. In at least one embodiment, the ribs **224** can extend from the wall **222** of the flux collar support section **220** of the valve body **200** along one or more axes perpendicular to a plane at which two or more sections **410** of the flux collar **400** interface (which can, but need not, include contacting one another).

[0035] In at least one embodiment, the flux collar support section **220** of the valve body **200** can include a wall **222** having an annular inner wall or surface **222a** and an outer wall or surface **222b**. In at least one embodiment, the outer wall **222b** of the flux collar support section **220** can have one or more at least partially flat or linear sides **226** (see, e.g., FIGS. **4** and **6**). In at least one embodiment, the wall **222** of the flux collar support section **220** can include one or more lobes **228** (see, e.g., FIGS. **4-6**). In at least one embodiment, the inner perimeter of the flux collar **400** can be an inner wall **402** having one or more at least partially flat or linear sides **406** corresponding to the sides **226** of the outer wall **222b** of the flux collar support section **220** of the valve body **200**. In at least one embodiment, the inner perimeter of the flux collar **400** can be an inner wall **402** having one or more contours or nonlinear portions corresponding to, or matching, the outer wall **222b** of the flux collar support section **220** of the valve body **200**.

[0036] In at least one embodiment, the flux collar support section **220** of the valve body **200** can have a plurality of longitudinal ribs **224** that form at least a portion of the interior surface of bore **240**, whether separately or in combination with wall **222** or one or more sections or portions thereof, and the flux collar **400** can have an inner wall **402** with a plurality of longitudinal grooves **404** configured to receive at least a portion of corresponding ones of the ribs **224**. For example, in at least one embodiment, such as a valve wherein at least a portion of the bore **240** is fluidically isolated from media flowing through the valve, one or more of the ribs **224** and the flux washer **400** can extend into or form a portion of the bore **240** (see, e.g., FIG. **7**). In such an embodiment, which is but one of many, the inner wall **402** of the flux collar **400** can be (but need not be) flush with an inner wall of the bore **240**, which can include being flush with an inner wall of the coil support section **230** of the valve body **200** or, for instance, being flush with an inner wall of the bore **240** along the flux collar support section **220**.

[0037] In at least one embodiment, the coil support section **230** of the valve body **200** can include an annular wall **232** having a thickness, such as a constant thickness along a length of the coil support section **230**. In at least one embodiment, the flux collar support section **220** of the valve body **200** can include an annular wall **222** having a variable thickness, such as a thickness that varies around or about the wall **222** (e.g., in a radial direction) and/or along a length of the flux collar support section **220**. In at least one embodiment, at least a portion of the variable thickness of the wall **222** of the flux collar support section **220** (such as the major or minor thickness of the wall **222**) can be less than the thickness of the coil support section **230** around or about at least a portion of the flux collar support section **220** of the valve body **200**.

[0038] In at least one embodiment, a wall **222** of the flux collar support section **220** of the valve body **200** can have a variable thickness, which can vary around the flux collar support section **220** of the valve body **200**. In at least one embodiment, the coil support section **230** of the valve body **200** can have a constant thickness. In at least one embodiment, a maximum thickness of the wall **222** of the flux collar support section **230** of the valve body **200** can be less than the thickness of the wall **232** of the coil support section **230** of the valve body **200** around at least a portion of the

flux collar support section **220**. In at least one embodiment, at least a portion of the variable thickness of the wall **222** of the flux collar support section **220** of the valve body **200** can be thicker than at least a portion of the thickness of the wall **232** of the coil support section **230** of the valve body **200**, such as along at least a portion of the flux collar support section **220**.

[0039] In at least one embodiment, valve body **200** can include one or more port sections **210**, one or more flux collar support sections **220**, one or more coil support sections **230**, or any combination thereof. Two or more of the sections **210**, **220**, **230** can be disposed along or can extend along one or more axes, such as a longitudinal axis (e.g., a central longitudinal axis). In at least one embodiment, the port section **210**, the flux collar support section **220**, and the coil support section **230** of the valve body **200** can be portions of a single, unitary body, such as an injection molded body. In at least one embodiment, the flux collar support section **220** of the valve body **200** can have a variable thickness, such as a wall thickness that varies around or about at least a portion of the flux collar support section **220** of the valve body **200**.

[0040] In at least one embodiment, the wall **232** of the bore **240** of the valve body **200** can have a constant thickness along at least a portion of the coil support section **230** of the valve body **200**. In at least one embodiment, the variable thickness of the wall **222** of the bore **240** along the flux collar support section **220** of the valve body **200** can be thinner than the thickness of the wall **232** of the bore **240** along the coil support section **230** of the valve body **200**, such as along at least a portion of the flux collar support section **220**. In at least one embodiment, the variable thickness of the wall **222** of the bore **240** along the flux collar support section **220** of the valve body **200** can be thicker than the wall **232** of the bore **240** along the coil support section **230** of the valve body **200**, such as around or about at least a portion of the flux collar support section **220**.

[0041] In at least one embodiment, a valve can include a valve body, a plunger within the valve body, a flux collar coupled to the valve body, or any combination thereof. In at least one embodiment, a distance between the flux collar and the plunger can vary around an inner perimeter of the flux collar. In at least one embodiment, the valve body can include a port section, a flux collar support section, and a coil support section extending along a longitudinal axis. In at least one embodiment, the port section, the flux collar support section, and the coil support section of the valve body can be a single monolithic structure, such as by way of being portions of a unitary or integrally formed structure. In at least one embodiment, the port section, the flux collar support section, and the coil support section of the valve body can be injection molded, such as out of plastic or another moldable material.

[0042] In at least one embodiment, the plunger can extend at least partially through the flux collar support section and the coil support section. In at least one embodiment, the flux collar can be coupled to the flux collar support section of the valve body. In at least one embodiment, the plunger can extend at least partially through the flux collar.

[0043] In at least one embodiment, the flux collar support section of the valve body can include an annular wall and/or a plurality of longitudinal ribs. In at least one embodiment, the ribs can extend from the annular wall of the flux collar support section of the valve body. In at least one embodiment, the inner perimeter of the flux collar can be an annular inner wall and/or can have a plurality of longitudinal grooves spaced to receive the ribs.

[0044] In at least one embodiment, the flux collar can be a multi-section ring. In at least one embodiment, each section of the flux collar can have at least one longitudinal groove configured to receive one of the ribs. For example, the flux collar can be a two-section ring. In at least one embodiment, the sections of the flux collar can interface along a plane. In at least one embodiment, one or more of the ribs can extend from the annular wall of the flux collar support section of the valve body, such as in a direction perpendicular to the plane.

[0045] In at least one embodiment, the flux collar support section of the valve body can have an annular inner wall and an outer wall. In at least one embodiment, the outer wall of the flux collar support section can have one or more flat sides and/or one or more lobes, such as one or more lobes

that extend radially outwardly relative to a longitudinal axis of the bore. In at least one embodiment, the inner perimeter of the flux collar can be an inner wall having at least two flat sides corresponding to the flat sides of the outer wall of the flux collar support section of the valve body. In at least one embodiment, the inner perimeter of the flux collar can be an inner wall having a contour corresponding to, or matching, the outer wall of the flux collar support section of the valve body.

[0046] In at least one embodiment, the flux collar support section of the valve body can have a plurality of longitudinal ribs, with or without an annular wall. In at least one embodiment, the inner perimeter of the flux collar can be an annular inner wall having a plurality of longitudinal grooves spaced to receive the ribs. In at least one embodiment, the inner wall of the flux collar can be annular and can be flush with an inner wall of the coil support section of the valve body.

[0047] In at least one embodiment, the coil support section of the valve body includes an annular wall having a constant thickness. In at least one embodiment, the flux collar support section of the valve body includes an annular wall having a variable thickness, which can vary around the wall. In at least one embodiment, the variable thickness of the flux collar support section can be less than the constant thickness of the coil support section around at least a portion of the flux collar support section of the valve body.

[0048] In at least one embodiment, the flux collar support section of the valve body can have a variable thickness, that varies around the flux collar support section of the valve body. In at least one embodiment, the coil support section of the valve body can have a constant thickness. In at least one embodiment, the variable thickness of the flux collar support section of the valve body can be thinner than the constant thickness of the coil support section of the valve body around at least a portion of the flux collar support section. In at least one embodiment, the variable thickness of the flux collar support section of the valve body can be thicker than the constant thickness of the coil support section of the valve body around at least a portion of the flux collar support section.

[0049] In at least one embodiment, valve body can include a port section, a flux collar support section, and a coil support section extending along a longitudinal axis, such as a central longitudinal axis or another axis. In at least one embodiment, the port section, the flux collar support section, and the coil support section of the valve body can be unitary. In at least one embodiment, the flux collar support section of the valve body can have a variable thickness, such as a wall thickness that varies around at least a portion of the flux collar support section of the valve body.

[0050] In at least one embodiment, the coil support section of the valve body can have a constant thickness around the coil support section of the valve body. In at least one embodiment, the variable thickness of the flux collar support section of the valve body can be thinner than the thickness of the coil support section of the valve body around at least a portion of the flux collar support section. In at least one embodiment, the variable thickness of the flux collar support section of the valve body can be thicker than the thickness of the coil support section of the valve body around at least a portion of the flux collar support section.

[0051] Other and further embodiments utilizing one or more aspects of the disclosure can be devised without departing from the spirit of Applicants' disclosure. For example, the devices, systems and methods can be implemented for numerous different types and sizes in numerous different industries. Further, the various methods and embodiments of the devices, systems and methods can be included in combination with each other to produce variations of the disclosed methods and embodiments. Discussion of singular elements can include plural elements and vice versa. The order of steps can occur in a variety of sequences unless otherwise specifically limited. The various steps described herein can be combined with other steps, interlineated with the stated steps, and/or split into multiple steps. Similarly, elements have been described functionally and can be embodied as separate components or can be combined into components having multiple functions.

[0052] The inventions have been described in the context of preferred and other embodiments and not every embodiment of the inventions has been described. Obvious modifications and alterations to the described embodiments are available to those of ordinary skill in the art having the benefits of the present disclosure. The disclosed and undisclosed embodiments are not intended to limit or restrict the scope or applicability of the inventions conceived of by the Applicants, but rather, in conformity with the patent laws, Applicants intend to fully protect all such modifications and improvements that come within the scope or range of equivalents of the following claims.

Claims

1. A valve comprising: a valve body having a port section, a flux collar support section, and a coil support section extending along a longitudinal axis; a plunger within the valve body and extending at least partially through the flux collar support section and the coil support section; and a flux collar mounted on the flux collar support section of the valve body; wherein the plunger extends at least partially through the flux collar; and wherein a distance between the flux collar and the plunger varies around an inner perimeter of the flux collar.
2. The valve of claim 1, wherein the flux collar support section of the valve body comprises an annular wall having a plurality of longitudinal ribs.
3. The valve of claim 2, wherein the ribs extend radially outwardly from the annular wall of the flux collar support section of the valve body.
4. The valve of claim 2, wherein the inner perimeter of the flux collar comprises an annular inner wall having a plurality of longitudinal grooves spaced to receive the ribs.
5. The valve of claim 4, wherein the port section, the flux collar support section, and the coil support section of the valve body comprise a unitary structure; wherein the flux collar comprises a multi-section ring; and wherein each section of the flux collar includes at least one longitudinal groove configured to receive one of the ribs.
6. The valve of claim 5, wherein the flux collar comprises a two-section ring; wherein the sections of the flux collar interface along a plane; and wherein the ribs extend from the annular wall of the flux collar support section of the valve body in a direction perpendicular to the plane.
7. The valve of claim 1, wherein the flux collar support section of the valve body comprises an annular inner wall and an outer wall having at least two flat sides.
8. The valve of claim 7, wherein the inner perimeter of the flux collar comprises an inner wall having at least two flat sides corresponding to the flat sides of the outer wall of the flux collar support section of the valve body.
9. The valve of claim 1, wherein the flux collar support section of the valve body comprises an annular inner wall and a plurality of lobes extending radially outwardly therefrom.
10. The valve of claim 9, wherein the inner perimeter of the flux collar comprises an inner wall having a contour matching a contour of an outer wall of the flux collar support section of the valve body.
11. The valve of claim 1, wherein the flux collar support section of the valve body comprises a plurality of longitudinal ribs.
12. The valve of claim 11, wherein an inner perimeter of the flux collar comprises an annular inner wall having a plurality of longitudinal grooves spaced to receive the ribs.
13. The valve of claim 12, wherein the annular inner wall of the flux collar is flush with an inner wall of the coil support section of the valve body.
14. The valve of claim 1, wherein the coil support section of the valve body comprises an annular wall having a constant thickness; wherein the flux collar support section of the valve body comprises an annular wall having a variable thickness; and wherein the variable thickness of the flux collar support section is less than the constant thickness of the coil support section around at least a portion of the flux collar support section of the valve body.

15. The valve of claim 1, wherein the flux collar support section of the valve body has a variable thickness that varies around the flux collar support section of the valve body.

16. The valve of claim 15, wherein the coil support section of the valve body has a constant thickness; and wherein the variable thickness of the flux collar support section of the valve body is thinner than the constant thickness of the coil support section of the valve body around at least a portion of the flux collar support section.

17. The valve of claim 16, wherein the variable thickness of the flux collar support section of the valve body is thicker than the constant thickness of the coil support section of the valve body around at least a portion of the flux collar support section.

18. A valve body comprising: a port section, a flux collar support section, and a coil support section extending along a longitudinal axis; wherein the port section, the flux collar support section, and the coil support section of the valve body comprise a unitary injection molded structure; and wherein the flux collar support section of the valve body has a variable thickness that varies around the flux collar support section of the valve body.

19. The valve of claim 18, wherein the coil support section of the valve body has a constant thickness around the coil support section of the valve body; and wherein the variable thickness of the flux collar support section of the valve body is thinner than the constant thickness of the coil support section of the valve body around at least a portion of the flux collar support section.

20. The valve of claim 19, wherein the variable thickness of the flux collar support section of the valve body is thicker than the constant thickness of the coil support section of the valve body around at least a portion of the flux collar support section.
